

Frequency-Direction Aware Multichannel Selective Fixed-Filter Active Noise Control Based on Multi-Task Learning

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Zhengding Luo ; Dongyuan Shi ; Xiruo Su ; Woon-Seng Gan [All Authors](#)

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Abstract:

The selective fixed-filter active noise control (SFANC) method has gained popularity due to its high computational efficiency and fast response. However, it solely accounts for noise frequency information, while neglecting its spatial information, which undoubtedly undermines the noise reduction performance, especially for direction-varied noises. To address this limitation, we proposed the Frequency-Direction Aware Multichannel SFANC (FD-MCSFANC) method, which incorporates both frequency and direction information of primary noises for accurate filter selection in the multichannel ANC (MCANC) system. In our approach, a Convolutional Neural Network (CNN) is designed to classify noises based on both frequency components and Direction-of-Arrival (DOA), with the combined classification results determining selected control filters. Furthermore, a joint loss function based on multi-task learning is utilized to implement end-to-end training of the CNN. Numerical simulations show that the FD-MCSFANC method effectively attenuates noises with different frequencies and incident angles. Moreover, it responds much faster than traditional adaptive algorithms and achieves better global noise control performance than the Multichannel SFANC (MCSFANC) method.

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SECTION I. Introduction

Noise pollution has become one of the top environmental hazards with the increased use of industrial equipment [1], [2]. Noise control strategies can be broadly categorized into Passive Noise Control (PNC) and Active Noise Control (ANC) [3], [4]. The PNC techniques aim at attenuating unwanted noises by using absorbers or barriers [5], [6]. However, the implementation of PNC methods to handle low-frequency noises is costly and makes noise control systems bulky [7], [8]. As a complementary technique to PNC, ANC is considered an effective method for reducing low-frequency noise [9], [10], [11]. It works by creating a sound field that cancels unwanted sounds based on the principle of superposition [12], [13].

Traditional ANC systems usually employ adaptive algorithms to adjust control filter coefficients, aiming to minimize the squared error signal [14]. The most common adaptive algorithms are the filtered-reference least mean square (FxLMS) algorithm and its variations [15], [16]. Although adaptive algorithms can cope with changes in noise source and environment, they are limited by slow convergence speed and the possibility of divergence [17], [18], [19]. To enhance convergence speed and stability, many commercial products have adopted fixed-filter ANC methods, in which control filters are pre-trained without adaptive updates [20].

However, fixed-filter ANC methods are typically effective only for specific types of noise and may result in poor performance for other noise types [21]. To address this limitation, deep learning techniques with powerful learning capabilities have emerged as promising solutions to enhance ANC performance [22], [23], [24], [25], [26]. Recent research has introduced Selective Fixed-Filter Active Noise Control (SFANC) methods based on deep learning [27], [28], [29], where Convolutional Neural Networks (CNNs) are employed to select pre-trained control filters for different primary noises. In these data-driven methods, the



filter selection mechanism is automatically learned from training data, without the need for prior knowledge or human settings [30].

Although these SFANC methods have proven effective for frequency-varied noises [31], they are limited to local noise control as single-channel ANC systems and do not account for the spatial information of primary noises. Local noise control around the error microphone is insufficient for larger areas, such as high-dimensional ventilation ducts [32], automobiles [33], and open windows [34]. Multichannel ANC (MCANC) is widely recognized as an effective solution to enlarge the noise reduction area [35]. In this paper, we employ a multichannel SFANC (MCSFANC) system to achieve a larger noise-cancellation zone. Moreover, since SFANC methods rely solely on noise frequency components to select control filters, their performance will degrade when dealing with noises coming from different directions [36], [37]. Some research [38], [39] find that feedforward ANC systems are vulnerable to noises with different Direction-of-Arrival (DOA). Therefore, in addition to frequency information, we incorporate direction information into the MCSFANC method, rendering it both frequency- and direction-aware.

Some methods [40], [41], [42], [43], [44] have introduced DOA estimation techniques in ANC systems, as summarized in Table I. However, they did not use direction information for control filter selection. Additionally, these ANC methods utilized conventional DOA estimation techniques [45], [46], [47]. Recent research in DOA estimation has highlighted the superiority of deep learning over conventional techniques in terms of accuracy, robustness, and adaptability [48], [49]. Therefore, this paper proposes a Frequency-Direction Aware Multichannel SFANC (FD-MCSFANC) method based on multi-task learning. We use a lightweight CNN to respectively classify noises based on their frequency information and direction information. A joint loss function is employed for end-to-end training of the CNN via multi-task learning. By integrating the results of frequency and direction classifications, a suitable control filter can be selected.

TABLE I Comparison of the Proposed FD-MCSFANC Method With Some Related ANC Methods

Approach	DOA estimation	ANC method	Key features
Hase et al. [40]	Cross-correlation	FxLMS	Select reference microphones that satisfy causality constraint
Liu et al. [41]	GCC and SRP	FxLMS	Select a reference microphone close to noise source
Patel et al. [42]	Beamforming	Leaky FxLMS	A directional hear-through ANC headphone system
Zhang et al. [43]	Beamforming	FxLMS	Reproduce sound from the desired direction
Xiao et al. [44]	Beamforming	Constrained LMS	Preserve the desired sound instead of reproducing it
Shi et al. [27]	-	Single-channel SFANC via CNN	Select filters based on frequency information
FD-MCSFANC	CNN	Multichannel SFANC via CNN	Select filters based on frequency and direction information

As an enhancement of the SFANC method, the FD-MCSFANC method can select appropriate control filters for noises with different frequency components and DOAs. Especially, the selection of filters is automatically handled by the CNN trained through multi-task learning. Moreover, unlike the local noise control of the single-channel SFANC method, the FD-MCSFANC method can achieve a larger noise control area. In real-time noise control, a co-processor runs the CNN to select appropriate control filters for each noise frame, while the real-time controller simultaneously processes noise on a sample-by-sample basis. This coordinated operation between the two units allows for delayless noise control.

The structure of this paper is as follows: Section II covers the fundamentals of MCANC systems and DOA estimation based on Time Difference of Arrival (TDOA). Section III describes the proposed FD-MCSFANC method in detail. Numerical simulations evaluating the method's effectiveness are presented in Section IV. Finally, Section V concludes this article.

SECTION II. Preliminaries

This section presents a basic introduction to feedforward MCANC systems and DOA estimation using TDOA. The limitations of adaptive MCANC algorithms and traditional TDOA estimation techniques are also discussed.

A. Feedforward MCANC System

MCANC systems are widely utilized in complex acoustic environments to enlarge the noise-cancellation zone. MCANC systems generate control signals given the reference signals to drive the corresponding secondary sources and use error sensors to measure the residual noise components. Fig. 1 illustrates the block diagram of an MCANC system, which consists of J reference microphones, K secondary sources, and M error microphones. The reference signal vector picked up by the j -th ($j \in \{1, 2, \dots, J\}$) reference microphone is given by

$$\mathbf{x}_j(n) = [x_j(n) \quad x_j(n-1) \quad \cdots \quad x_j(n-L_f+1)]^T, \quad (1)$$

[View Source](#)

where L_f denote the length of the control filter, and T represent the matrix transpose operation.

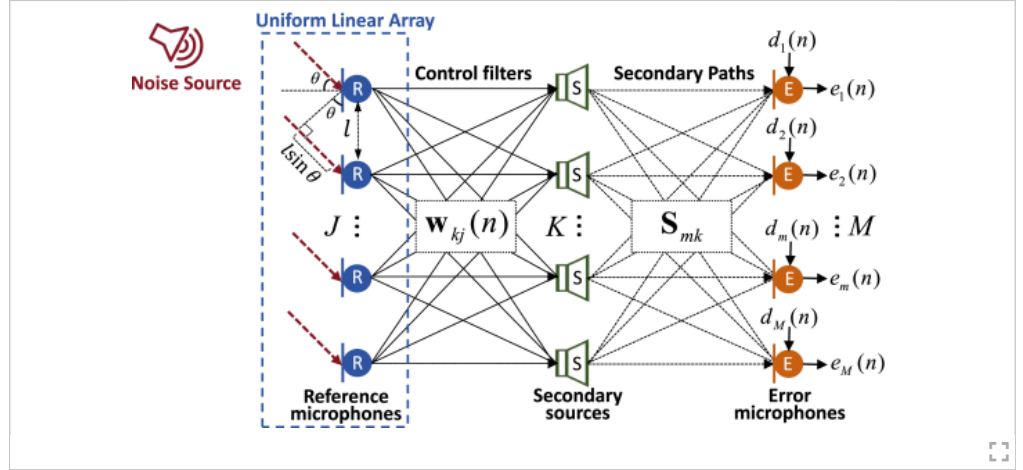


Fig. 1.

The schematic of the feedforward MCANC system with J reference microphones, K secondary sources, and M error microphones, where J reference microphones with an inner distance l form a uniform linear array.

The control signal utilized to drive the k -th secondary source is obtained from

$$y_k(n) = \sum_{j=1}^J \mathbf{x}_j^T(n) \mathbf{w}_{kj}(n), \quad k = 1, 2, \dots, K, \quad (2)$$

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where $\mathbf{w}_{kj}(n)$ denotes the control filter vector from the j -th reference microphone to the k -th secondary source and can be articulated as

$$\mathbf{w}_{kj}(n) = [w_{kj,1}(n) \quad w_{kj,2}(n) \quad \dots \quad w_{kj,L_f}(n)]^T. \quad (3)$$

[View Source](#)

The error signal picked up by the m -th error microphone is obtained from

$$e_m(n) = d_m(n) - \sum_{k=1}^K \mathbf{y}_k^T(n) \mathbf{s}_{mk} \quad m = 1, 2, \dots, M, \quad (4)$$

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in which $d_m(n)$ denotes the disturbance at the m -th error microphone, and $\mathbf{s}_{mk}$ represents the impulse response of the secondary path from the k -th secondary source to the m -th error microphone, denote as

$$\mathbf{s}_{mk} = [s_{mk,1} \quad s_{mk,2} \quad \dots \quad s_{mk,L_s}]^T, \quad (5)$$

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where L_s denotes the length of the impulse response of the secondary path. The control signal vector $\mathbf{y}_k(n)$ is given by

$$\mathbf{y}_k(n) = [y_k(n) \quad y_k(n-1) \quad \dots \quad y_k(n-L_s+1)]^T. \quad (6)$$

[View Source](#)

In traditional adaptive MCANC systems, according to the Multichannel FxLMS (MCFxLMS) [3] algorithm, the update equation of the control filter is computed as

$$\mathbf{w}_{kj}(n+1) = \mathbf{w}_{kj}(n) + \mu \sum_{m=1}^M \mathbf{x}'_{jkm}(n) e_m(n), \quad (7)$$

[View Source](#)

where μ denotes the step size. $\mathbf{x}'_{jkm}(n)$ stands for the filtered reference signal generated by passing the reference signal through the estimate of the secondary path $\hat{s}_{mk}(n)$ as

$$\mathbf{x}'_{jkm}(n) = \hat{s}_{mk}(n) * \mathbf{x}_j(n), \quad (8)$$

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in which, $*$ represents the linear convolution operator.

It is worth noting that inappropriate step sizes will significantly affect traditional adaptive MCANC systems. The step size adjustment depends on the trade-off between convergence speed and stability, where a more significant step size may cause instability, and a smaller step size may result in slow convergence [50]. To achieve both a rapid response and a high level of stability, suitable pre-trained control filters will be selected for different primary noises in the proposed FD-MCSFANC method.

B. Time Difference of Arrival (TDOA)

Fig. 1 illustrates an MCANC system, which employs a Uniform Linear Array (ULA) of reference microphones with a spacing of l between adjacent microphones. The primary noise source is assumed to originate from the far field at an incident angle of θ . Consequently, the relative time delay between the j -th reference microphone and the first reference microphone can be expressed as

$$\tau_j = \frac{(j-1)l \sin \theta}{c}, \quad j = 1, 2, \dots, J, \quad (9)$$

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where c denotes the sound velocity in air, and the angle θ ranges between -90° and 90° . If τ_j is known, then θ can be uniquely determined, and vice versa. Consequently, τ_j is commonly known as the time difference of arrival (TDOA) in far-field scenarios and is usually used to determine the incident angle θ of the noise wave [51].

Traditional methods for TDOA estimation include cross-correlation, generalized cross-correlation (GCC), multiple signal classification (MUSIC), and adaptive eigenvalue decomposition (AED). However, these conventional approaches often depend on strong assumptions about signal and noise characteristics, which may not hold in complex real-world scenarios [52], [53]. In contrast, deep neural networks are capable of learning intricate non-linear relationships between input signals and their corresponding TDOA values directly from data, leading to improved localization performance in challenging environments [48]. Accordingly, in our FD-MCSFANC method, a CNN is developed to automatically learn the TDOA information of primary noises via a data-driven mechanism. Additionally, unlike traditional methods that focus solely on estimating direction, the FD-MCSFANC approach leverages multi-task learning to jointly perform both frequency and direction classification, improving noise control performance and computational efficiency.

SECTION III.

The Proposed FD-MCSFANC Method

Previous SFANC methods [20], [27], [28], [31] primarily relied on noise frequency components to select control filters, but their performance tends to degrade when dealing with noises with different DOAs. In this paper, we propose an FD-MCSFANC approach that incorporates both frequency and direction information of primary noises to select appropriate control filters, as illustrated in Fig. 2. Inspired by multi-task learning

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Block diagram of the proposed FD-MCSFANC approach, where a co-processor executes the CNN model to perform frequency and direction classifications, followed by selecting control filters for each noise frame, while real-time noise control is executed at the sampling rate in parallel.

Obtaining pre-trained control filters is an essential step in the proposed FD-MCSFANC method. In this phase, the MCANC system depicted in Fig. 1 is utilized to cancel $F \cdot T$ types of synthetic band-limited noises. These synthetic band-limited noises have F different frequency bands and T different arriving angles. The MCFxLMS algorithm is employed to derive optimal control filters for these band-limited noises, as described in Equation (7). Consequently, a set of pre-trained control filters, denoted as $\{\mathbf{W}_{y,z}\}_{y=1,z=1}^{F,T}$ and consisting of $F \cdot T$ filters, is stored in the control filter database.

In this subsection, we detailed the architecture of the CNN, the dataset used for its training, and the CNN training process via multi-task learning.

The CNN plays a crucial role in selecting control filters for various primary noises in the FD-MCSFANC approach. The architecture of the CNN, detailed in Table II, is a lightweight model adapted from ShuffleNet V2 [55], which is known for its classification efficiency and accuracy. The initial input consists of a J -channel reference signal, which is transformed into J magnitude spectrograms and J phase spectrograms using the Short-Term Fourier Transform (STFT). These STFT spectrograms, forming a total of $2 \cdot J$ channels, are then fed into the CNN.

Layer	Output Size	Output Channels
Input	(64, 126)	$2 \cdot J$
Stage1: Shared representation		
Conv0	(64, 126)	10
Conv1	(64, 126)	3
Conv2	(32, 63)	24
MaxPool	(16, 32)	24
Frequency classification		
Stage2	(8, 16)	48
Stage3	(4, 8)	96
AdaptiveAvgPool	(1, 1)	96
FC	-	F
Direction classification		
Stage2	(8, 16)	48
Stage3	(4, 8)	96
AdaptiveAvgPool	(1, 1)	96
FC	-	T

In the CNN, the input STFT spectrograms are first processed by the Stage1 module. Two convolutional layers (Conv0 and Conv1) transform the $2 \cdot J$ -channel input into a 3-channel tensor to match the input size of the original ShuffleNet V2. The Conv2 layer and a max-pooling layer then produce a shared feature representation of size $24 \times 16 \times 32$ for frequency and direction classifications. In the frequency classification and direction classification modules, Stage2 and Stage3 are the same as those in ShuffleNet V2, consisting of convolutional layers, batch normalization, and ReLU activation layers. Additionally, the fully connected (FC) layer is configured with F outputs for frequency classification and T outputs for direction classification, respectively.

2) Training Dataset

A synthetic noise dataset comprising N instances is used to train the CNN for classifying primary noises based on their frequency components and DOAs. Each instance, denoted as $\mathbf{x}_i$, contains J channels, and each channel has a length of L_r . The process for generating these instances is depicted in Fig. 3. First, N single-channel band-limited noises with randomly selected center frequencies and bandwidths are generated. Each single-channel band-limited noise is then treated as the first-channel signal and expanded into a corresponding J -channel broadband noise by applying a random incident angle θ . The time delay τ_j between the j -th channel and the first channel is calculated using (9). The frequency and direction labels for these instances are determined by comparing their frequency bands and incident angles with those of the filter-training broadband noises, as described in Section III-A.

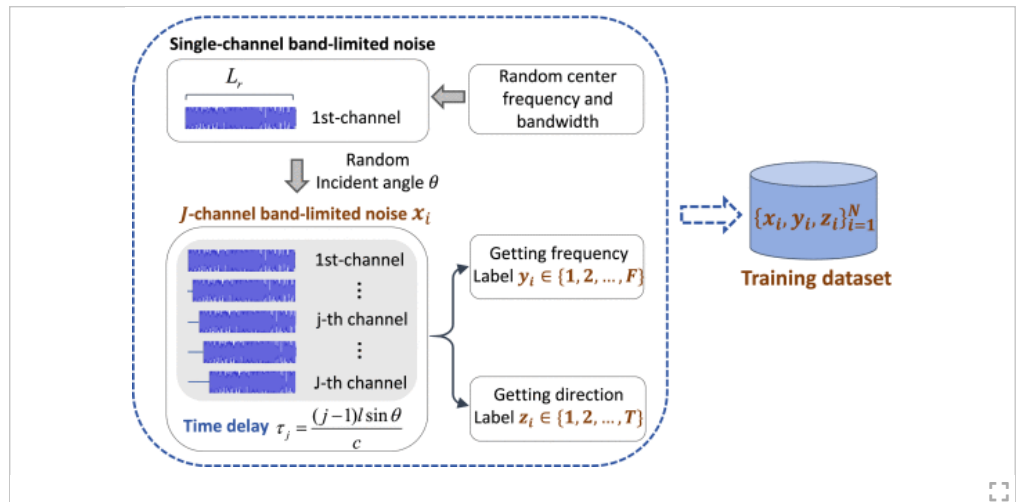


Fig. 3.

The dataset used to train the CNN consists of N noise instances, each being J -channel, with corresponding frequency and direction labels.

3) Multi-Task Learning

The process of end-to-end CNN training based on multi-task learning is shown in Fig. 4(a). The noise dataset $\{\mathbf{x}_i, y_i, z_i\}_{i=1}^N$ has been generated as introduced in Section III-B2, where

- $\mathbf{x}_i \in \mathbb{R}^{J \times L_r}$ denotes the i -th noise instance composed of J channels, each channel with a length of L_r . $\mathbb{R}$ represents the set of real numbers.
- $y_i \in \{1, 2, \dots, F\}$ represents the frequency label for $\mathbf{x}_i$, where F is the number of frequency categories.
- $z_i \in \{1, 2, \dots, T\}$ represents the direction label for $\mathbf{x}_i$, where T is the number of direction categories.

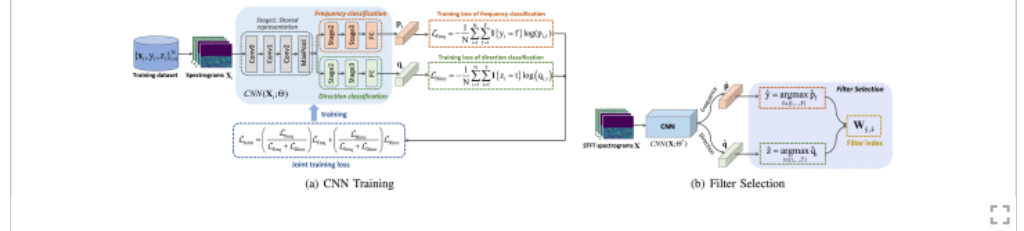


Fig. 4.

(a) End-to-end training of the CNN based on multi-task learning; (b) Filter selection process using the trained CNN.

As introduced in Section III-B1, the noise instance $\mathbf{x}_i$ is transformed into the STFT spectrograms $\mathbf{X}_i$ as the input of the network. The proposed CNN, parameterized by Θ , maps the input $\mathbf{X}_i$ to the predicted probabilities for each frequency class and direction class:

$$\begin{aligned} (\mathbf{p}_i, \mathbf{q}_i) &= \text{CNN}(\mathbf{X}_i; \Theta), \\ \mathbf{p}_i &= [p_{i,1} \dots p_{i,f} \dots p_{i,F}] \in \mathbb{R}^{1 \times F}, \\ \mathbf{q}_i &= [q_{i,1} \dots q_{i,t} \dots q_{i,T}] \in \mathbb{R}^{1 \times T}, \end{aligned} \quad (10)$$

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where $\text{CNN}(\cdot)$ denotes the operations in the CNN. $\mathbf{p}_i$ and $\mathbf{q}_i$ represent the predicted probability vector over all frequency categories and direction categories, respectively. $p_{i,f}$ is the predicted probability that $\mathbf{X}_i$ belongs to the f -th frequency category, and $q_{i,t}$ is the predicted probability that $\mathbf{X}_i$ belongs to the t -th direction category.

During training the CNN, the training losses of direction classification and frequency classification are separately expressed as

$$\begin{aligned} \mathcal{L}_{\text{Freq}} &= -\frac{1}{N} \sum_{i=1}^N \sum_{f=1}^F 1\{y_i = f\} \log(p_{i,f}), \\ \mathcal{L}_{\text{Dir}} &= -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^T 1\{z_i = t\} \log(q_{i,t}), \end{aligned} \quad (11)$$

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where $1(\cdot)$ is the indicator function, which equals 1 if the condition inside is true and 0 otherwise. $\mathcal{L}_{\text{Freq}}$ and $\mathcal{L}_{\text{Dir}}$ are computed using cross-entropy loss function. By integrating $\mathcal{L}_{\text{Freq}}$ and $\mathcal{L}_{\text{Dir}}$, a joint loss function used for end-to-end training of the CNN is given by

$$\mathcal{L}_{\text{Joint}} = \left(\frac{\mathcal{L}_{\text{Freq}}}{\mathcal{L}_{\text{Freq}} + \mathcal{L}_{\text{Dir}}} \right) \mathcal{L}_{\text{Freq}} + \left(\frac{\mathcal{L}_{\text{Dir}}}{\mathcal{L}_{\text{Freq}} + \mathcal{L}_{\text{Dir}}} \right) \mathcal{L}_{\text{Dir}}. \quad (12)$$

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The joint loss is a weighted sum of the individual losses for each task, with the weight of each task's loss dynamically adjusted based on the CNN's performance in both tasks. These dynamic weights enable the

network to focus more on the task that is currently more challenging, promoting more balanced performance improvements across the two tasks.

The training process aims to minimize the joint loss $\mathcal{L}_{\text{Joint}}$ with respect to the CNN parameters Θ , using optimizers such as SGD, Adam, etc. Optimization is performed iteratively, with Θ being updated in each iteration to reduce the loss until a stopping criterion is reached. Ultimately, the optimal parameters of the CNN are obtained as

$$\Theta^* = \underset{\Theta}{\operatorname{argmin}} \mathcal{L}_{\text{Joint}}. \quad (13)$$

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It is worth noting that the joint loss based on multi-task learning leverages the interdependencies between frequency classification and direction classification to enhance performance and efficiency. By training the network on both tasks simultaneously, it learns a shared representation that improves generalization [54], [56]. Moreover, joint training acts as a form of regularization, as the network must perform well on both tasks, which helps prevent overfitting to a single task. Additionally, training the CNN jointly is often more efficient than training separate networks for each task, both in terms of computational resources and time [57].

C. Filter Selection

After training, the CNN loads its optimal parameters Θ^* to perform frequency classification and direction classification. The results from the two tasks are combined to select a control filter from the pre-trained control filter database $\{\mathbf{W}_{y,z}\}_{y=1,z=1}^{T,F}$. Block diagram of the filter selection process is shown in Fig. 4(b). Given a frame of the reference signal $\mathbf{x}$, the CNN takes its STFT spectrogram $\mathbf{X}$ as input to predict the probabilities for each frequency class and direction class:

$$\begin{cases} (\hat{\mathbf{p}}, \hat{\mathbf{q}}) = \text{CNN}(\mathbf{X}; \Theta^*), \\ \hat{\mathbf{p}} = [\hat{p}_1 \dots \hat{p}_f \dots \hat{p}_F] \in \mathbb{R}^{1 \times F}, \\ \hat{\mathbf{q}} = [\hat{q}_1 \dots \hat{q}_t \dots \hat{q}_T] \in \mathbb{R}^{1 \times T}, \end{cases} \quad (14)$$

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where $\hat{\mathbf{p}}$ and $\hat{\mathbf{q}}$ are the predicted probability vectors for frequency classification and direction classification, respectively. $\hat{p}_f$ is the predicted probability that the input signal belongs to the f -th frequency category. $\hat{q}_t$ is the predicted probability that the input signal belongs to the t -th direction category.

Subsequently, the selected pre-trained control filter $\hat{\mathbf{W}}$ for the reference signal $\mathbf{x}$ is determined by

$$\begin{cases} \hat{\mathbf{W}} = \mathbf{W}_{\hat{y}, \hat{z}} \in \{\mathbf{W}_{y,z}\}_{y=1,z=1}^{F,T}, \\ \hat{y} = \underset{f \in \{1, \dots, F\}}{\operatorname{argmax}} \hat{p}_f, \\ \hat{z} = \underset{t \in \{1, \dots, T\}}{\operatorname{argmax}} \hat{q}_t, \end{cases} \quad (15)$$

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where $\hat{y}$ and $\hat{z}$ represent the indices of the maximum probabilities in $\hat{\mathbf{p}}$ and $\hat{\mathbf{q}}$, respectively. Consequently, $\hat{\mathbf{W}}$ is the selected control filter based on the indices $\hat{y}$ and $\hat{z}$, determined by both the frequency classification and direction classification results.

In summary, the filter selection process integrates both frequency and direction information, enabling the selection of suitable pre-trained control filters to attenuate noises with different frequencies and arrival angles. Additionally, the filter selection in the FD-MCSFANC method is entirely data-driven, requiring no additional prior knowledge or manual effort.

D. Delayless Noise Control

Throughout the noise control process, the real-time processor executes noise control on a sample-by-sample basis. Simultaneously, for each noise frame, the co-processor employs the trained CNN to perform frequency classification and direction classification. By combining the results of the two tasks, an appropriate control filter is selected from the pre-trained control filter database. The initial coefficients of the control filter are set to zero. If a different control filter is selected, it will replace the currently used control filter for subsequent noise control. Table III provides the pseudo-code for the noise control process in the FD-MCSFANC method.

TABLE III Pseudo-Code of Real-Time Noise Control in the FD-MCSFANC Method

Algorithm Description: The filter selection is performed in the co-processor at the frame rate, while real-time noise control is conducted at the sampling rate in parallel. Input: Each frame of the reference signal is denoted as $\mathbf{x}$. Note: The pre-trained control filter database is denoted as $\{\mathbf{W}_{y,z}\}_{y=1,z=1}^{F,T}$, each control filter $\mathbf{W}_{y,z} \in \mathbb{R}^{K \times J L_f}$. The initial coefficients of the used control filter $\mathbf{W}$ are set to zero, $\mathbf{W} = \mathbf{0}$. $STFT(\cdot)$ represents obtaining the STFT spectrograms. $CNN(\cdot)$ denotes the operations in the CNN, parameterized by Θ^*. While FD-MCSFANC on # Delayless noise control in the real-time controller: for each sample of the reference signal do $y_k(n) = \sum_{j=1}^J \mathbf{w}_{kj}^T(n) \mathbf{x}_j(n)$ ▷ Control signal used to drive the k-th secondary source. $e_m(n) = d_m(n) - \sum_{k=1}^K y_k(n) s_{mk}$ ▷ Error signal picked up by the m-th error microphone. end for # Control filter selection in the co-processor: for each frame of the reference signal do $\mathbf{X} \leftarrow STFT(\mathbf{x})$ ▷ STFT spectrograms of the reference signal. $(\hat{\mathbf{p}}, \hat{\mathbf{q}}) \leftarrow CNN(\mathbf{X}; \Theta^*)$ ▷ Frequency classification and direction classification. $\hat{y} = \underset{f \in \{1, \dots, F\}}{\operatorname{argmax}} \hat{p}_f, \hat{z} = \underset{t \in \{1, \dots, T\}}{\operatorname{argmax}} \hat{q}_t, \mathbf{W}' = \mathbf{W}_{\hat{y}, \hat{z}}$ ▷ Select a control filter based on classification results. # Update the used control filter in the real-time controller: if $\mathbf{W} \neq \mathbf{W}'$ do ▷ If the currently used control filter is different from the selected one. $\mathbf{W} \leftarrow \mathbf{W}'$ ▷ Update the used control filter for subsequent noise reduction. end for	
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In practical deployment, the FD-MCSFANC method allows a co-processor (e.g., a laptop, mobile phone, or cloud server) to run the CNN at the frame rate, while the real-time controller simultaneously performs noise cancellation at the sampling rate. This dual-rate architecture ensures that complex processing is handled by the co-processor without affecting the real-time performance of the ANC system, thereby enabling delayless noise control [58]. Moreover, the FD-MCSFANC method does not depend on error signals for control filter updates, leading to faster convergence and better stability compared to adaptive MCANC algorithms. This method provides a promising solution for handling noises of varied frequencies arriving from different directions in practical scenarios.

SECTION IV.

Numerical Simulations

In this section, the effectiveness of the proposed FD-MCSFANC method is evaluated through some numerical simulations.

A. Simulation Setup

Table IV summarizes the values of the variables used in the numerical simulations. The FD-MCSFANC system includes 4 reference microphones, 2 secondary sources, and 4 error microphones. Notably, error microphones are only used to obtain the pre-trained control filters and are not utilized during real-time noise control.

TABLE IV The Values of Variables in the Simulations

Variable	Definition	Value
J	Number of reference microphones	4
K	Number of secondary sources	2
M	Number of error microphones	4
l	Distance between adjacent reference microphones	0.24 m
c	Sound velocity in air	343 m/s
L_f	Number of control filter's taps	1024
L_s	Length of the impulse response of secondary path	512
-	Length of the impulse response of primary path	1024
F	Number of frequency categories	7
T	Number of direction categories	8
N	Number of noise instances in the training dataset	80000
-	Sampling rate	16000 Hz
L_r	Length of each channel in the J -channel noise instance	8000

B. Pre-Trained Control Filters

Initially, we synthesized 7 band-limited noises with frequency bands as shown in Fig. 5(a). These band-limited noises served as the primary noises, propagating to the 4 reference microphones from 8 different incident angles. As illustrated in Fig. 5(b), the incident angles of the primary noises ranged from -70° to 70° , with increments of 20° . Using the MCFxLMS algorithm, we derived 56 pre-trained control filters, each

corresponding to a specific combination of primary noise's frequency and direction. The indexes of pre-trained control filters, $\{\mathbf{W}_{y,z}\}_{y=1,z=1}^{7,8}$, are determined by both the frequency category and direction category, as shown in Fig. 5(c).

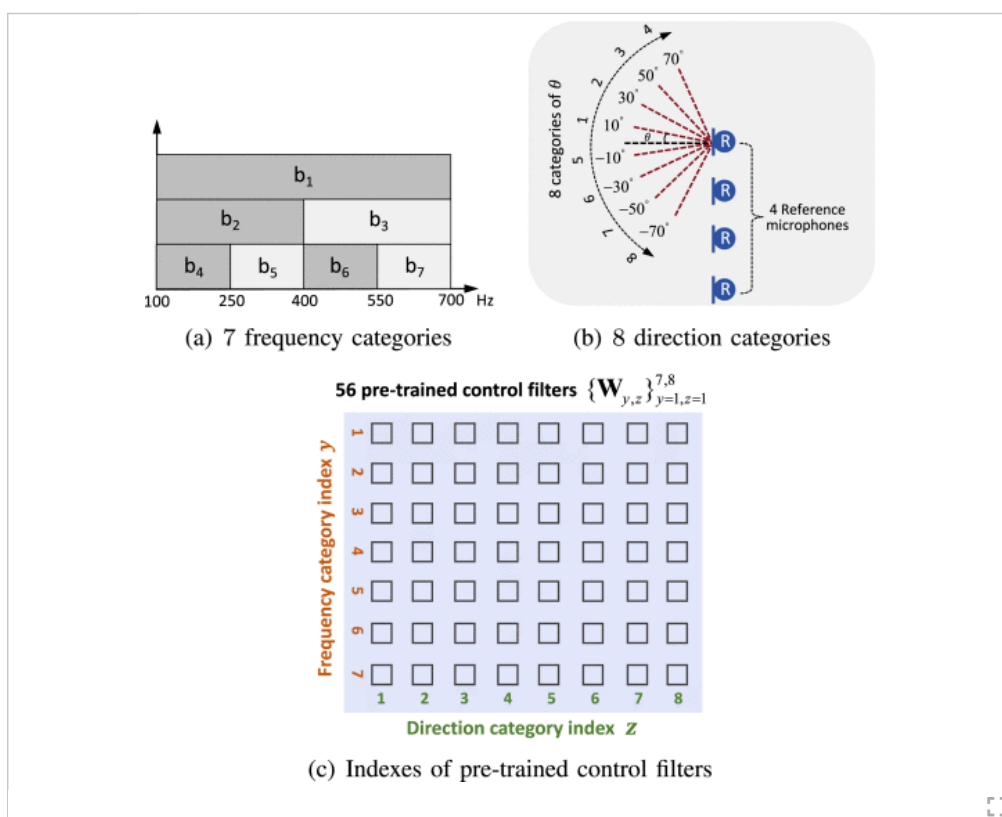


Fig. 5.

Different frequency bands in (a) correspond to 7 frequency categories. Different incident angles in (b) correspond to 8 direction categories. 56 pre-trained control filters in (c) correspond to different frequency and direction categories.

C. Effectiveness of the CNN

Noise data with various frequency bands and incident angles are used to train the CNN. After training, the model's capabilities in frequency classification and direction classification are evaluated on a testing dataset, and its computational complexity is also assessed.

1) Training Dataset

The CNN is trained using a synthetic noise dataset comprising 80,000 noise instances for training and 2,000 for testing, with each instance lasting 0.5 seconds and consisting of 4 channels. As described in Section III-B2, the generation of these noise instances involves three steps: (1) 80,000 single-channel band-limited noises are generated, with randomly selected center frequencies and bandwidths within 100–700 Hz. (2) Each single-channel noise is then expanded into a 4-channel noise corresponding to a random incident angle between -75° and 75° . (3) The closest frequency band and incident angle from the available 7 frequency bands and 8 incident angles, as shown in Fig. 5, are assigned as the frequency and direction labels, respectively.

2) Classification Accuracy

Using the synthetic dataset described above, the CNN is trained via multi-task learning to simultaneously perform frequency and direction classifications. As summarized in Table Y, the proposed CNN achieves a high accuracy of 98% for frequency classification and 99% for direction classification on the test dataset. This suggests that the CNN can accurately classify noises in both frequency and direction aspects. Additionally, the CNN model is highly efficient, utilizing only 0.11 million parameters and achieving a fast runtime of 8.52 milliseconds, meeting real-time processing requirements. With computational demands of 7.44 million MACs and 15.94 million FLOPs, the CNN’s complexity is relatively low for a deep learning model, making it well-suited for use in embedded systems or mobile devices. The input length of the CNN is one frame with a duration of 0.5 seconds.

TABLE V Performance of the CNN in FD-MCSFANC

Performance Metrics	The CNN
Frequency classification accuracy	98%
Direction classification accuracy	99%
Model parameters	0.11 million
Model runtime	8.52 milliseconds
Multiply-accumulate operations (MACs)	7.44 million
Floating-point operations (FLOPs)	15.94 million
Network input length	0.5 seconds

D. Noise Cancellation Performance

In this section, the FD-MCSFANC method is applied to attenuate broadband noises and real-world noises to evaluate its performance. The tested broadband noises and real noises are not included in the training dataset. The used acoustic paths are measured from an MCANC window. Since achieving a perfect plane wave in our experimental setup is challenging, we simulated the received plane wave by manually adding a TDOA between the 4-channel reference signals according to Equation (9). Since the performance at each error microphone is similar, we only present the results for the first error microphone. Notably, after the CNN is trained, its decisions depend on the frequency and direction information of the input noise, while the pre-trained control filters are specific to the acoustic paths. Therefore, when deploying the FD-MCSFANC method in different acoustic environments with the same reference microphone array configuration, only the pre-trained control filters need to be updated, while the CNN remains unchanged. However, if the reference microphone array is altered, the CNN needs to be re-trained.

1) Broadband Noise Cancellation

The FD-MCSFANC method is applied to cancel 56 different 4-channel broadband noises, with different frequency bands and incident angles. The noise reduction levels (NR) and the selected control filters for these broadband noises are summarized in Table VI. The results indicate that the FD-MCSFANC method selects appropriate pre-trained control filters for different broadband noises, owing to the CNN's high frequency and direction classification accuracies. The NR values highlight the effectiveness of the pre-trained control filters in attenuating these broadband noises. Additionally, error signals of the MCANC system using the FD-MCSFANC method for several example broadband noises are shown in Fig. 6. In Fig. 6, the averaged NR value is calculated every 0.5 seconds, unlike the overall NR value used in Table VI. Fig. 6 shows that the FD-MCSFANC method achieves good noise reduction performance and responds rapidly to these broadband noises.

TABLE VI Noise Reduction Levels in dB and Selected Control Filters Obtained by the FD-MCSFANC Method for Different Broadband Noises

Frequency	Direction	10°	30°	50°	70°	-10°	-30°	-50°	-70°
100-700 Hz		11.7 $W_{1,1}$	11.5 $W_{1,2}$	11.5 $W_{1,3}$	12.0 $W_{1,4}$	11.7 $W_{1,5}$	11.5 $W_{1,6}$	11.5 $W_{1,7}$	12.0 $W_{1,8}$
100-400 Hz		10.7 $W_{2,1}$	10.8 $W_{2,2}$	11.1 $W_{2,3}$	11.4 $W_{2,4}$	10.7 $W_{2,5}$	10.8 $W_{2,6}$	11.1 $W_{2,7}$	11.4 $W_{2,8}$
400-700 Hz		12.2 $W_{3,1}$	12.5 $W_{3,2}$	12.9 $W_{3,3}$	11.6 $W_{3,4}$	12.2 $W_{3,5}$	12.5 $W_{3,6}$	12.7 $W_{3,7}$	11.5 $W_{3,8}$
100-250 Hz		11.1 $W_{4,1}$	11.0 $W_{4,2}$	10.9 $W_{4,3}$	10.8 $W_{4,4}$	11.0 $W_{4,5}$	11.0 $W_{4,6}$	10.9 $W_{4,7}$	10.8 $W_{4,8}$
250-400 Hz		10.2 $W_{5,1}$	10.3 $W_{5,2}$	10.6 $W_{5,3}$	11.1 $W_{5,4}$	10.2 $W_{5,5}$	10.3 $W_{5,6}$	10.6 $W_{5,7}$	11.0 $W_{5,8}$
400-550 Hz		12.0 $W_{6,1}$	12.0 $W_{6,2}$	11.9 $W_{6,3}$	12.4 $W_{6,4}$	12.0 $W_{6,5}$	12.0 $W_{6,6}$	11.9 $W_{6,7}$	12.4 $W_{6,8}$
550-700 Hz		12.3 $W_{7,1}$	12.4 $W_{7,2}$	12.6 $W_{7,3}$	12.2 $W_{7,4}$	12.3 $W_{7,5}$	12.1 $W_{7,6}$	12.6 $W_{7,7}$	12.2 $W_{7,8}$

¹ Frequency and Direction represent frequency band and incident angle, respectively.
² Values and $W_{p,q}$ in the table refer to NR values and selected control filters.

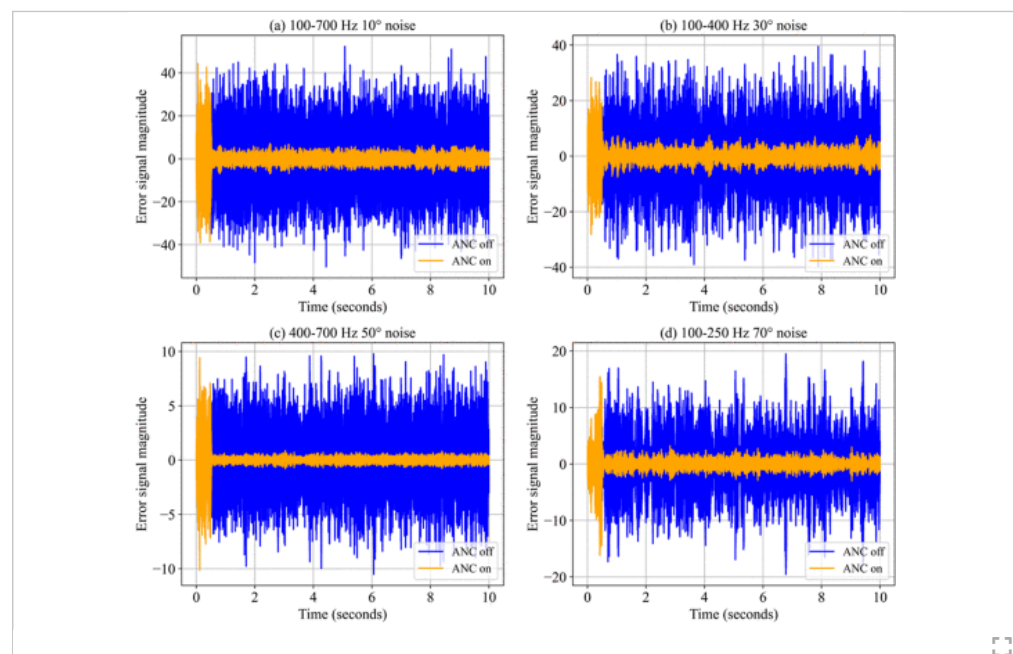
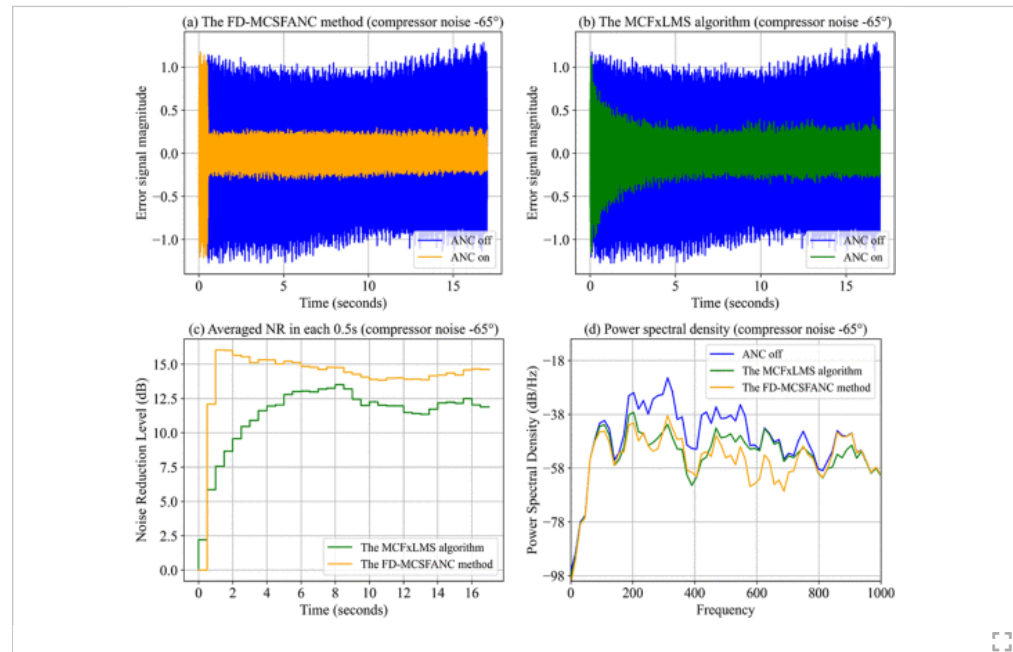


Fig. 6.

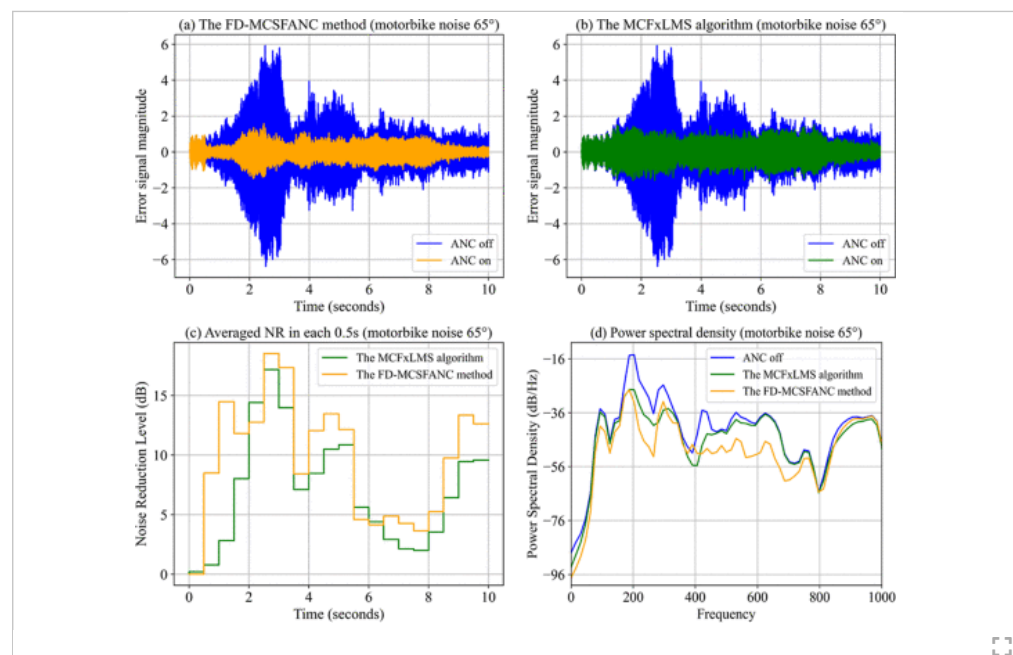
Error signals of the MCANC system using the FD-MCSFANC method for four different broadband noises.

2) Real Noise Cancellation

The performance of the FD-MCSFANC method is evaluated to handle a compressor noise with an incident angle of -65° and a motorbike noise with an incident angle of 65° . The frequencies and incident angles of these real noises fall outside the FD-MCSFANC's filter-training set. The performance of the FD-MCSFANC method on these noises is compared to that of the MCFxLMS algorithm, as shown in Figs. 7 and 8. The step size of the MCFxLMS algorithm is set to 1×10^{-7} . Since the CNN in FD-MCSFANC predicts the control filter every 0.5 s, we calculated the average NR value at this interval.

**Fig. 7.**

Noise reduction performance on the compressor noise with an incident angle of -65° , using the FD-MCSFANC method and the MCFxLMS algorithm.

**Fig. 8.**

Noise reduction performance on the motorbike noise with an incident angle of 65° , using the FD-MCSFANC method and the MCFxLMS algorithm.

The results in Figs. 7 and 8 demonstrate that the FD-MCSFANC method effectively attenuates both types of noise. Regarding the power spectral density figures, most frequency components between 100 Hz and 700 Hz of the two noises are reduced, aligning with the maximum frequency range of the pre-trained control filters. Moreover, the FD-MCSFANC method responds significantly faster to real-world noises compared to the MCFxLMS algorithm. For the compressor noise, the FD-MCSFANC method achieves an average NR value of about 15 dB after 0.5 seconds, while the MCFxLMS algorithm takes up to 8 seconds to reach comparable performance. When handling the motorbike noise, the NR values of the FD-MCSFANC method are higher than those of the MCFxLMS algorithm most of the time. These findings suggest that the MCFxLMS algorithm takes a long time to reach steady state, and its slow convergence significantly limits early-stage performance. Moreover, step size tuning is required to avoid divergence. In contrast, the FD-MCSFANC method responds rapidly and addresses the slow convergence issue inherent in adaptive algorithms. Additionally, the absence of feedback error signals during noise control in the FD-MCSFANC method minimizes the risk of divergence and enhances stability.

3) Varied Real-World Noise Cancellation

We sequentially cascade the 65° motorbike noise with 65° and -65° compressor noises, each lasting 10 seconds, to create a 30-second test signal. This signal is used to assess the effectiveness of the FD-MCSFANC method in controlling real-world noises with varied frequency components and incident angles. The performance results are shown in Fig. 9. The FD-MCSFANC method quickly adapts to changes in both frequency components and incident angles, requiring only 0.5 seconds to predict a new control filter when the noise characteristics change. It demonstrates the method's strong tracking ability to handle real-world noises' frequency and direction variations. In contrast, the MCFxLMS algorithm responds more slowly to noise variations due to its sample-by-sample updating process. Additionally, the FD-MCSFANC method achieves higher NR values than the MCFxLMS algorithm, particularly during the initial phases following each noise change. Thus, the varied real-world noise is dynamic and challenging for the MCFxLMS algorithm, while the FD-MCSFANC method proves more effective in handling it.

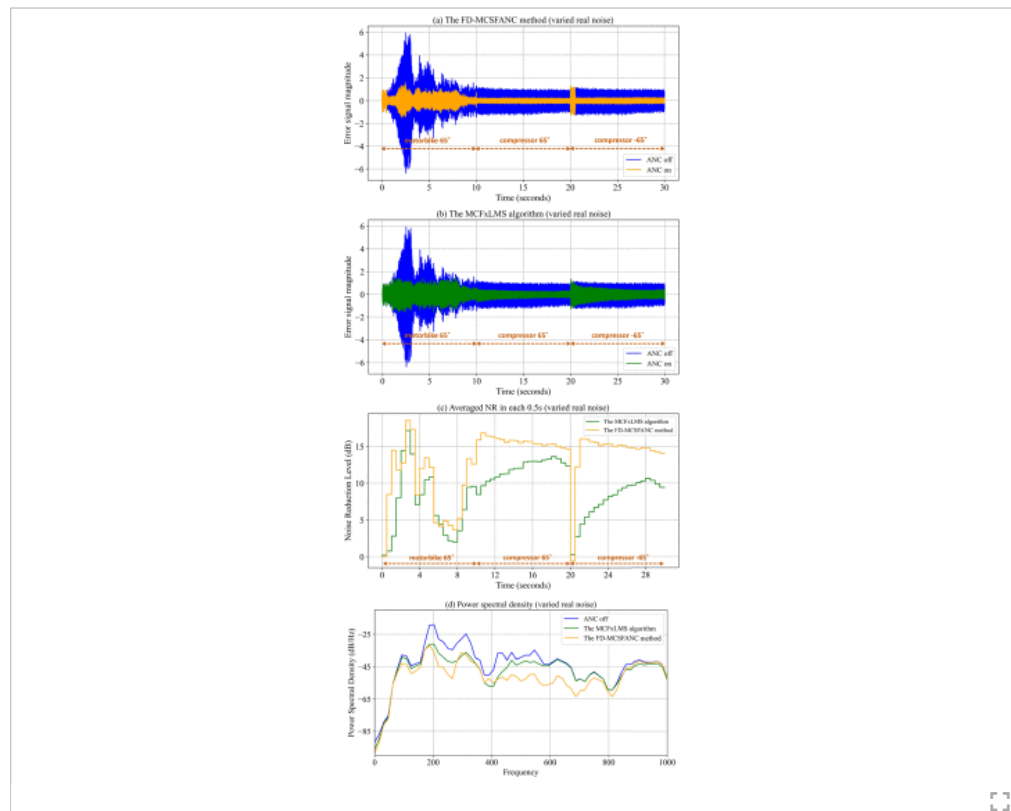


Fig. 9.

Error signals (a)–(b), averaged NR in each 0.5 seconds (c), and power spectral density (d), when using the FD-MCSFANC method and the MCFxLMS algorithm to attenuate the varied real-world noise.

E. Comparison With the MCSFANC Method

In this section, we compared the global control performance of the FD-MCSFANC and MCSFANC methods using simulations conducted with the Matlab k-Wave toolbox [59]. In the MCSFANC method [31], the incident angles of all noises are set to 0° by default, and control filters are selected based solely on their frequency information. In comparison, the proposed FD-MCSFANC method integrates the frequency and direction information of noises to select appropriate control filters. The simulation setup, illustrated in Fig. 10, consists of a far-field noise source, four reference microphones, two secondary sources, and four error microphones, forming a $4 \times 2 \times 4$ ANC system. Since the noise source is positioned at the end of the free field, the primary noise wave is regarded as a plane wave in this simulation. It is important to note that the 4

error microphones are used only during pre-training control filters and are not involved in the real-time noise control process.

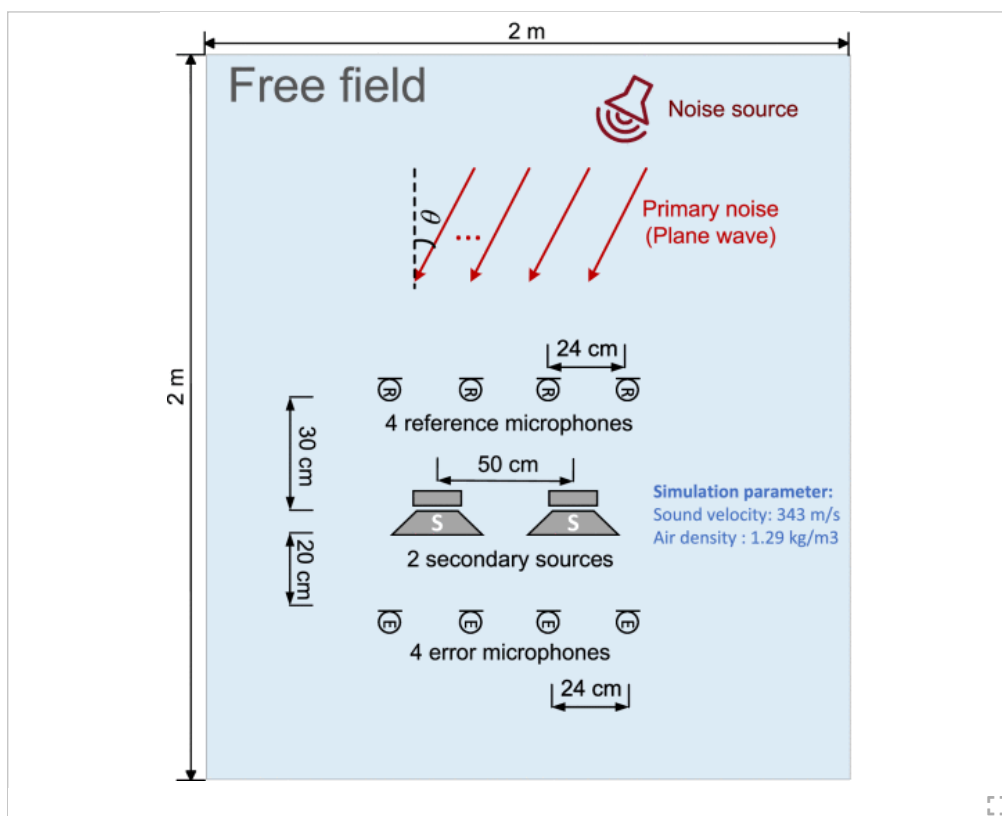
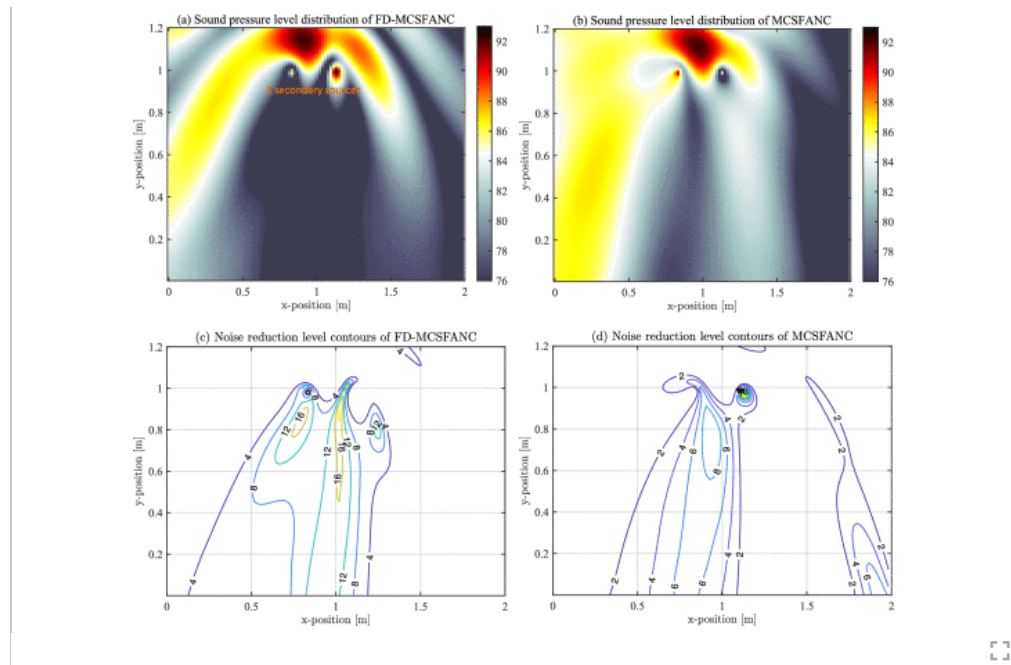


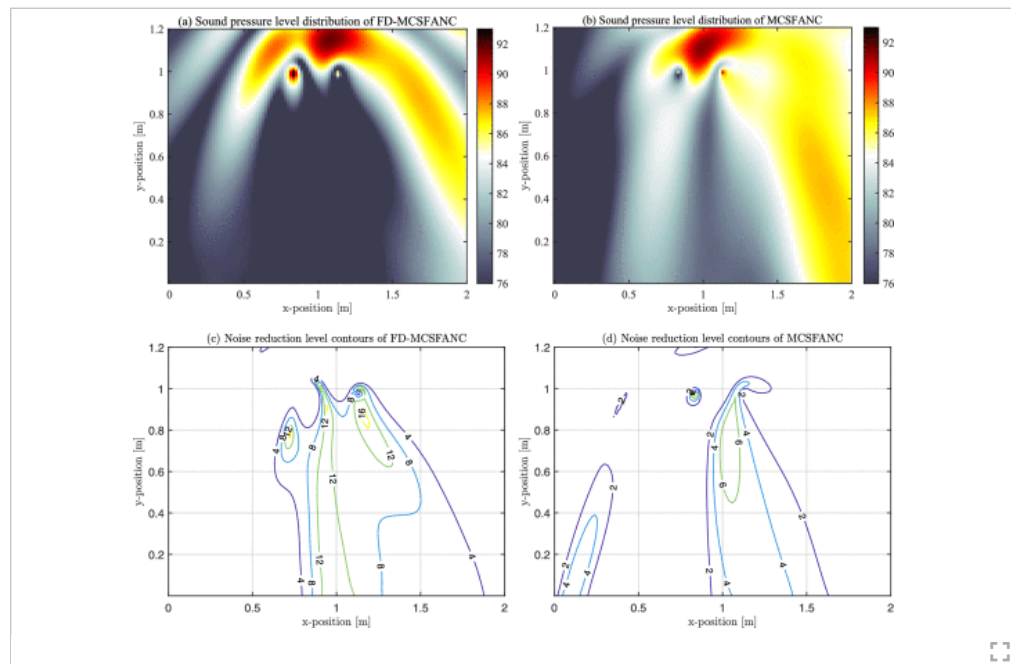
Fig. 10.

Settings of the $4 \times 2 \times 4$ MCANC system in the simulation platform using Matlab k-Wave [59].

Figs. 11 and 12 show sound pressure level distributions and NR contours of the FD-MCSFANC and MCSFANC methods when attenuating the $550\text{--}700\text{ Hz}$ noise with incident angles of 50° and -60° , respectively. The region behind the two secondary sources is seen as the zone of interest for validating the noise control performance. The results in Figs. 11 and 12 demonstrate that the FD-MCSFANC method is more effective in handling noises from various incident angles compared to the MCSFANC method. Regarding sound pressure level distributions, the FD-MCSFANC method achieves sound pressure levels approximately 6 dB lower than those obtained by the MCSFANC method. In terms of NR contours, the FD-MCSFANC method achieves $8\text{--}16\text{ dB}$, whereas the MCSFANC method reaches only about $4\text{--}8\text{ dB}$ in the zone of interest. Additionally, we find that the FD-MCSFANC method creates larger quiet zones than the MCSFANC method. This suggests that the MCSFANC method relies solely on frequency information to select control filters, making it difficult to handle noises arriving from different directions. The superiority of the FD-MCSFANC method is attributed to its ability to combine frequency and direction information to select suitable control filters, resulting in higher NR values and broader quiet zones.

**Fig. 11.**

Sound pressure level distributions (a)–(b) and NR contours (c)–(d) in dB obtained by the FD-MCSFANC and MCSFANC methods for the 550–700 Hz 50° noise.

**Fig. 12.**

Sound pressure level distributions (a)–(b) and NR contours (c)–(d) in dB obtained by the FD-MCSFANC and MCSFANC methods for the 550–700 Hz -60° noise.

SECTION V. Conclusion

This paper introduces the FD-MCSFANC method, a novel approach that enhances noise control performance by integrating both frequency and direction information. Unlike MCSFANC methods, which rely solely on the frequency information of primary noises, FD-MCSFANC utilizes both the frequency components and the DOA of primary noises to intelligently select pre-trained control filters. This filter selection process is automatically performed by a CNN trained using multi-task learning. Simulation results demonstrate that the FD-MCSFANC method effectively attenuates broadband noises and real-world noises with varied frequencies and incident angles while addressing the slow convergence issues associated with adaptive ANC algorithms.

Additionally, it provides improved global noise control performance compared to the MCSFANC method. A limitation of the current method is that the MCANC system employs a simple Uniform Linear Array configuration for the reference microphones. In future work, we will explore more complex array configurations to further evaluate and enhance the FD-MCSFANC method.

Authors	▼
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